

ADAUSDТ Multi-Horizon Forecast Report

Hybrid ARIMA + LightGBM + AutoEncoder + ResNet18

Test Period: 2026-01-01 to 2026-08-27

Horizons: 1 Hour, 1 Week, 1 Month

Metrics for 1H Horizon

	RMSE	MAE	MAPE	Directional_Accuracy	R2
arima	0.007841	0.005292	80884.75139	48.31735	-0.001477
lgbm	0.008055	0.005455	401707.327001	48.962511	-0.056715
resnet	0.010282	0.007684	1943110.790322	47.678275	-0.843636
fusion	0.007864	0.005313	174063.010503	49.206626	-0.007209
ae_lgbm	0.008201	0.005567	556086.362682	49.415867	-0.095511
naive	0.007836	0.005282	96.512642	3.487358	-9.9e-05

Metrics for 1W Horizon

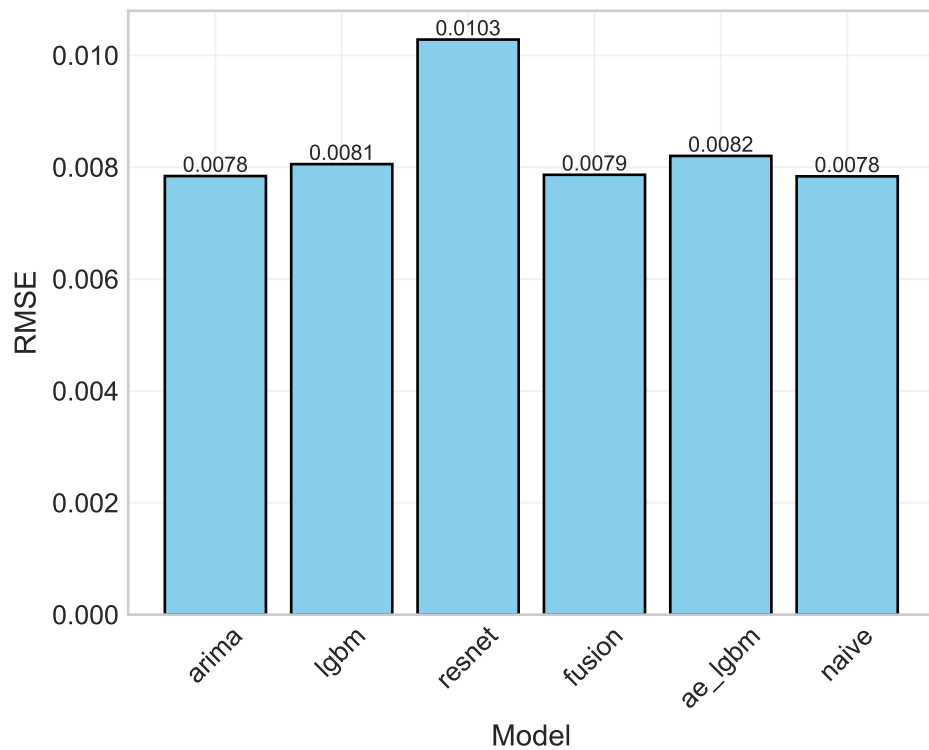
	RMSE	MAE	MAPE	Directional_Accuracy	R2
arima	0.108048	0.079496	16005.448053	47.593391	-0.026383
lgbm	0.19704	0.153352	5612961.916249	45.420259	-2.413395
resnet	0.102514	0.075626	1593063.686282	57.980929	-0.061316
fusion	0.141607	0.113795	2005469.614598	44.989224	-0.762984
ae_lgbm	0.163666	0.129415	4336433.382792	48.814655	-1.355025
naive	0.10808	0.079559	99.748563	0.251437	-0.026989

Metrics for 1M Horizon

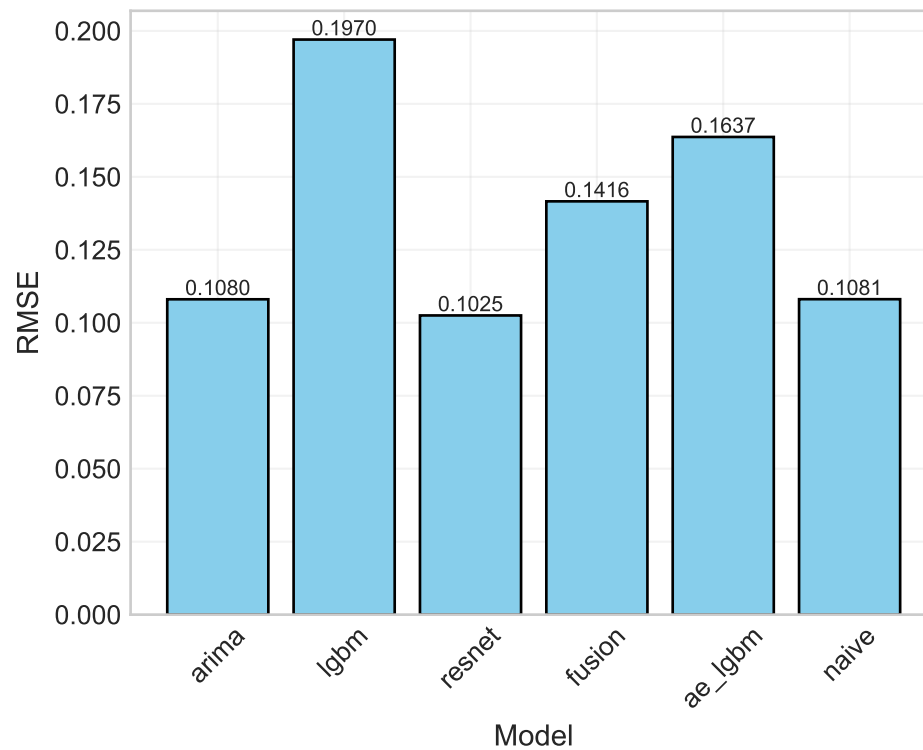
	RMSE	MAE	MAPE	Directional_Accuracy	R2
arima	0.229409	0.166168	12046.039246	46.19391	-0.235368
lgbm	0.396478	0.353315	4832994.374193	44.411058	-2.68987
resnet	0.226615	0.169436	1156188.807215	53.150912	-0.225796
fusion	0.264999	0.236302	3793726.695538	59.375	-0.648401
ae_lgbm	0.226533	0.173539	934629.215385	65.144231	-0.204582
naive	0.229451	0.16624	99.839744	0.160256	-0.235817

Comparison of RMSE Across Models and Horizons

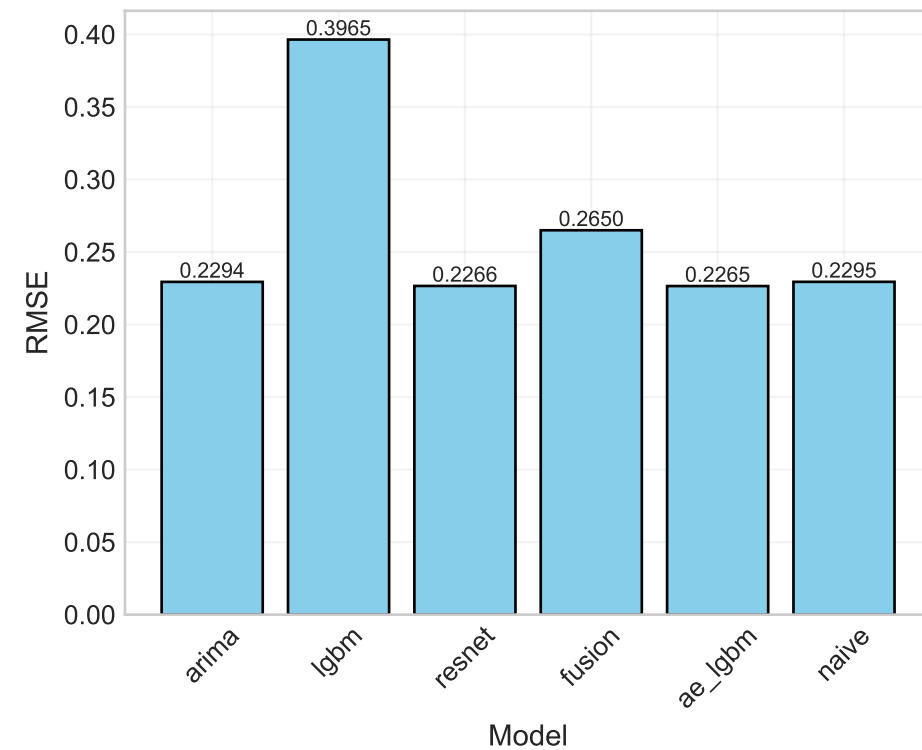
1H - RMSE



1W - RMSE

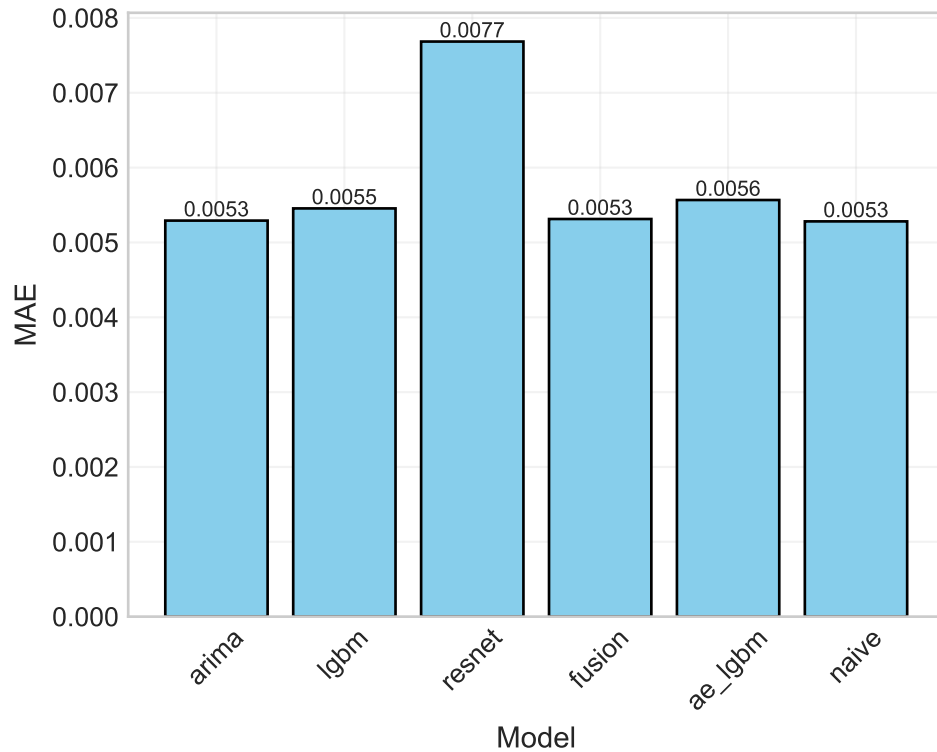


1M - RMSE

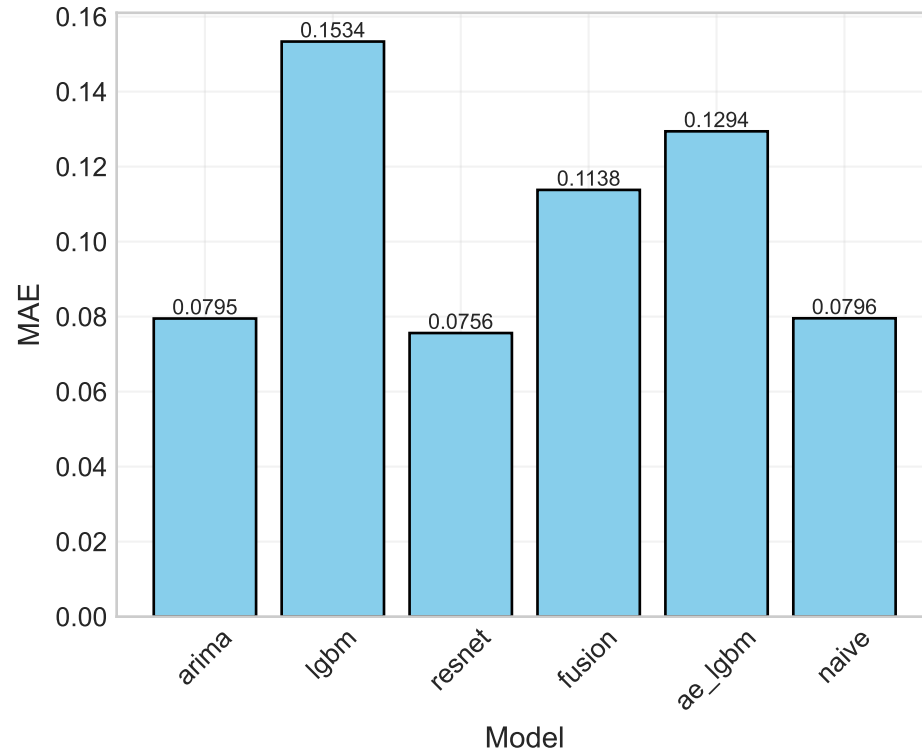


Comparison of MAE Across Models and Horizons

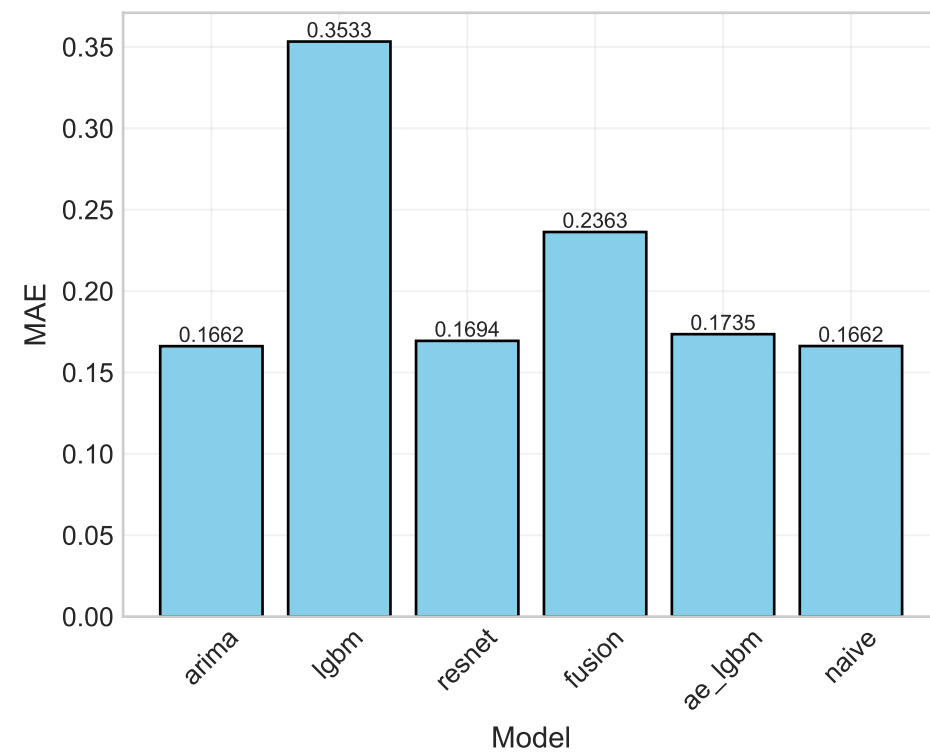
1H - MAE



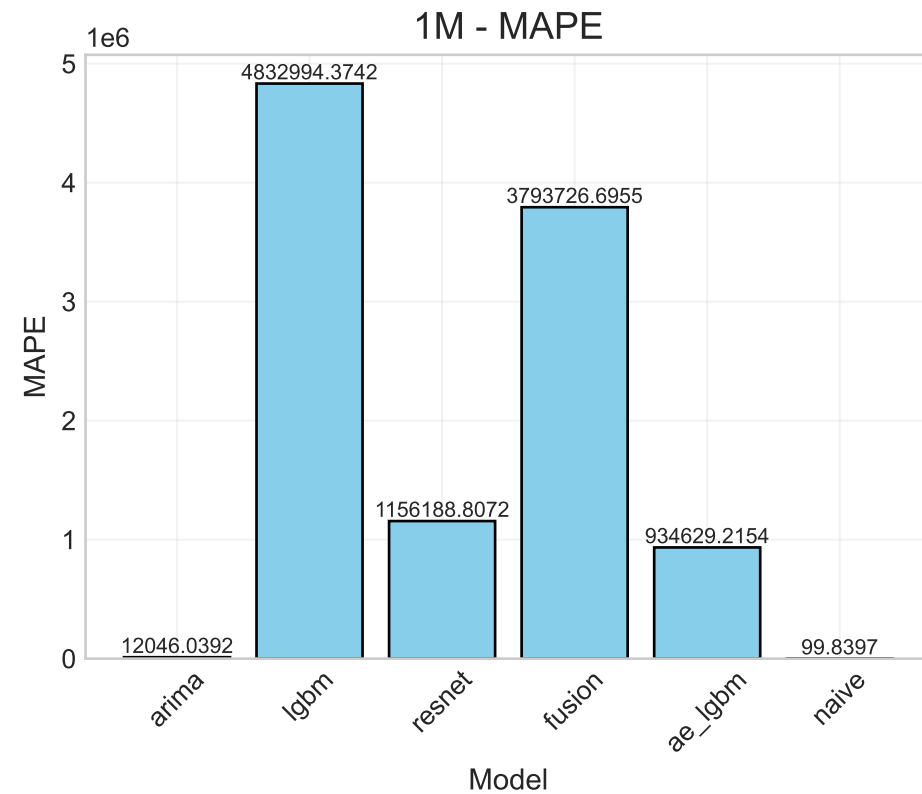
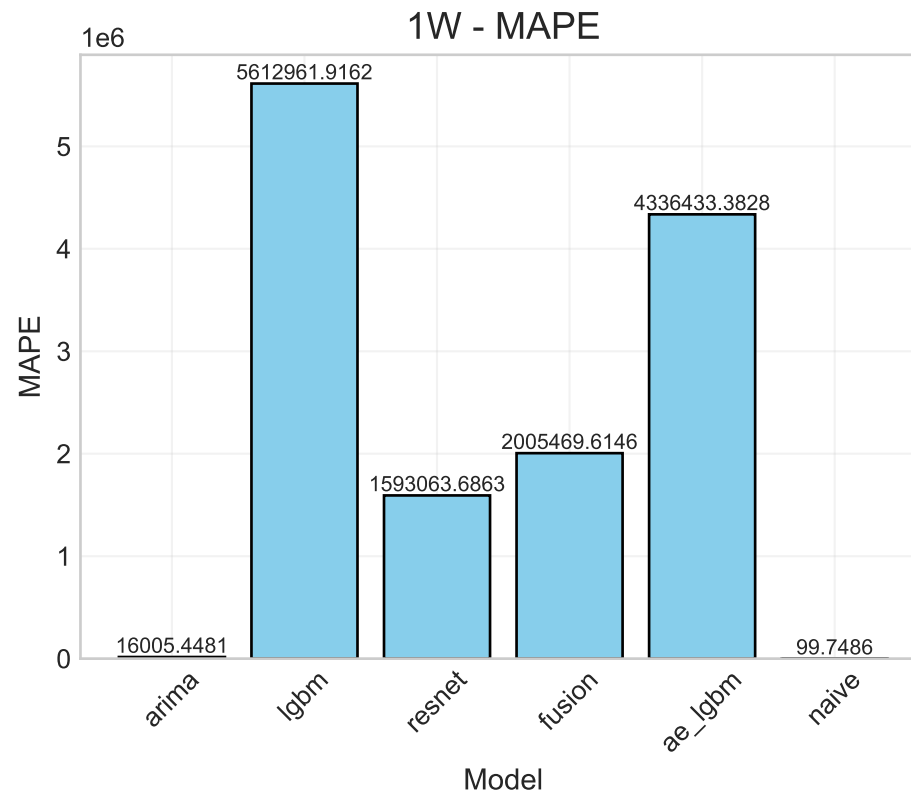
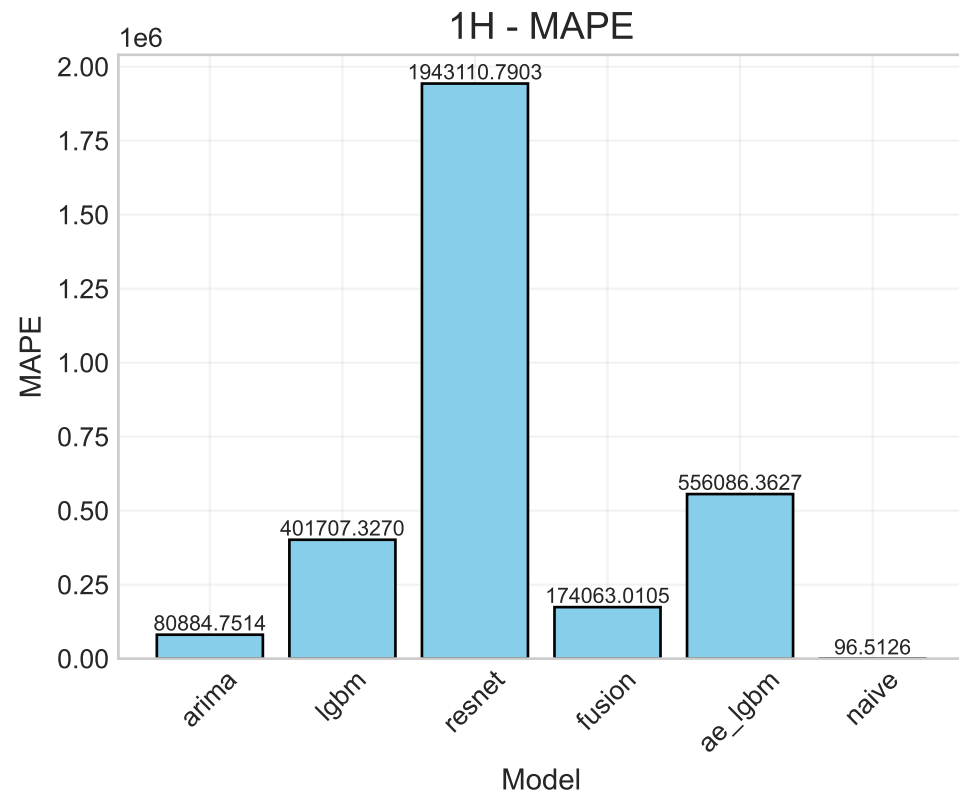
1W - MAE



1M - MAE

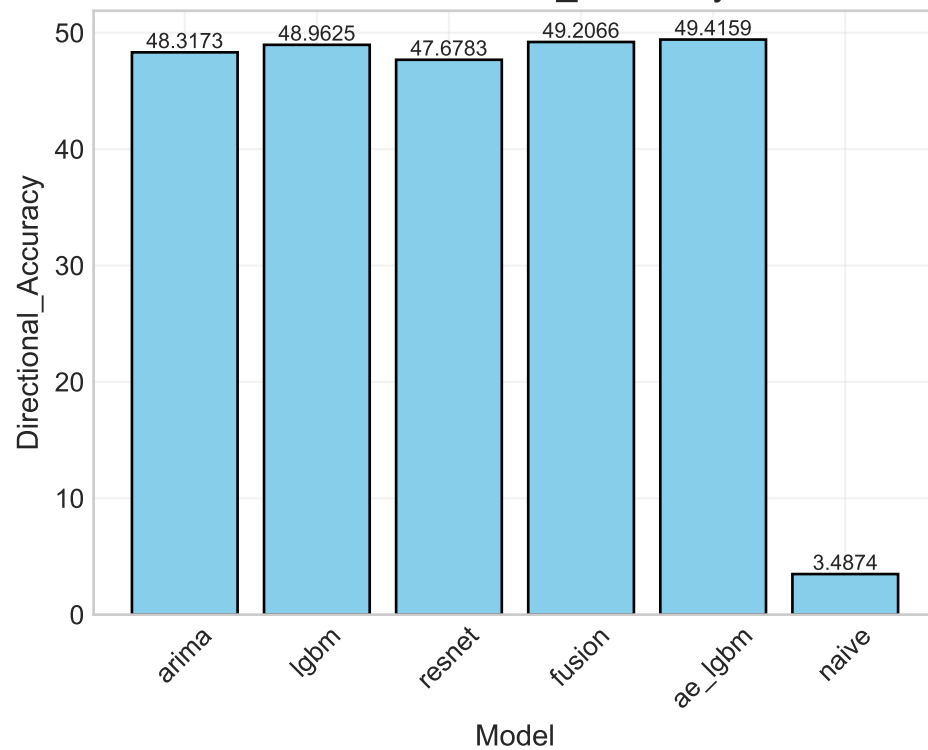


Comparison of MAPE Across Models and Horizons

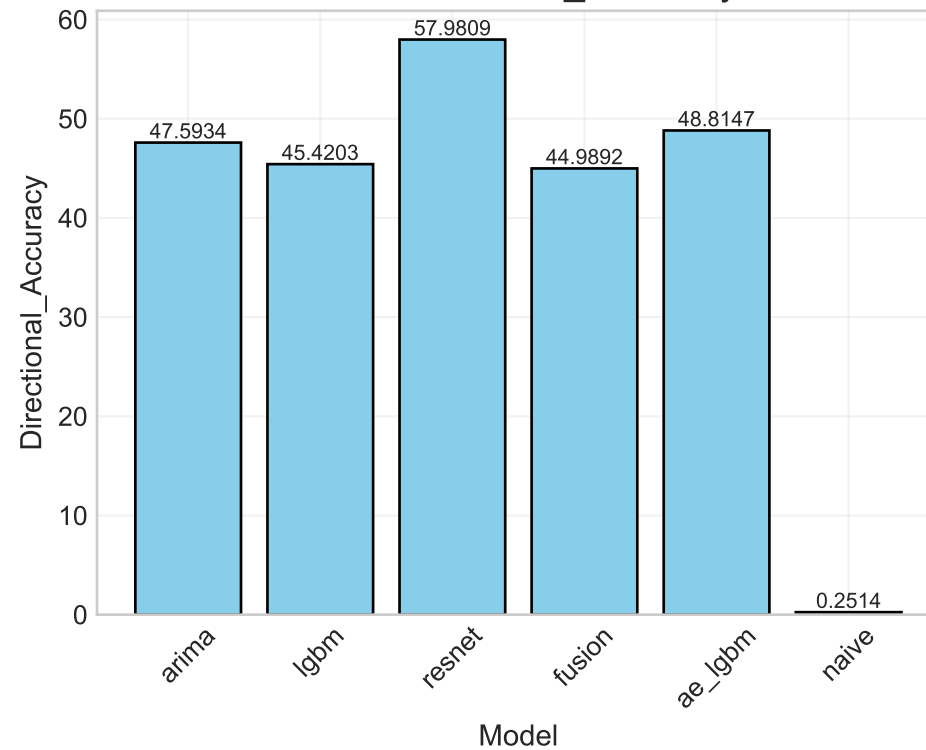


Comparison of Directional_Accuracy Across Models and Horizons

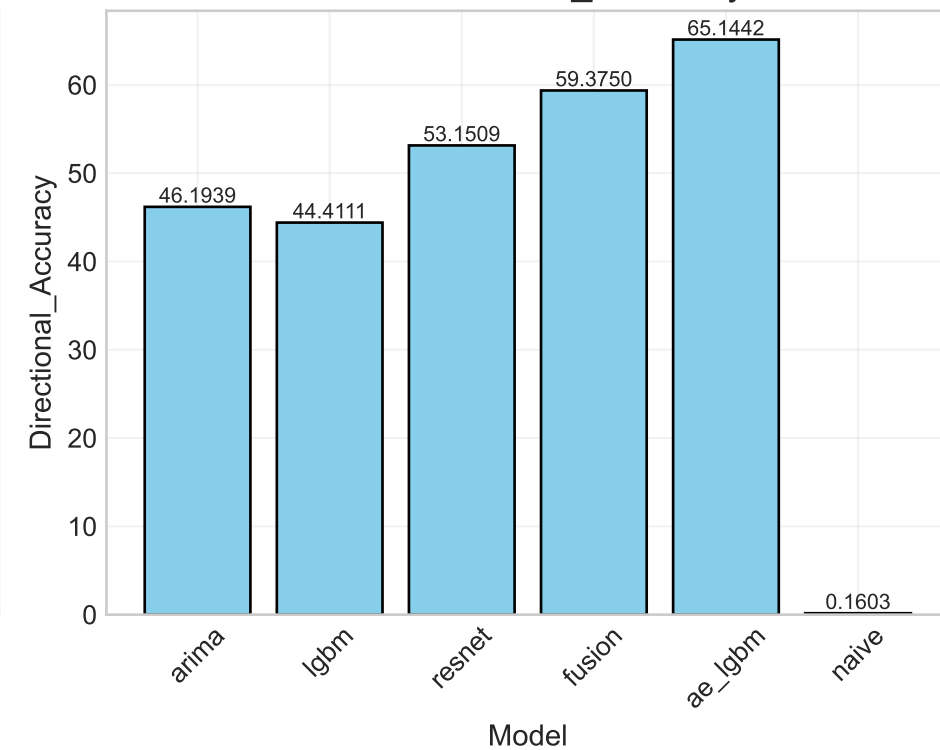
1H - Directional_Accuracy



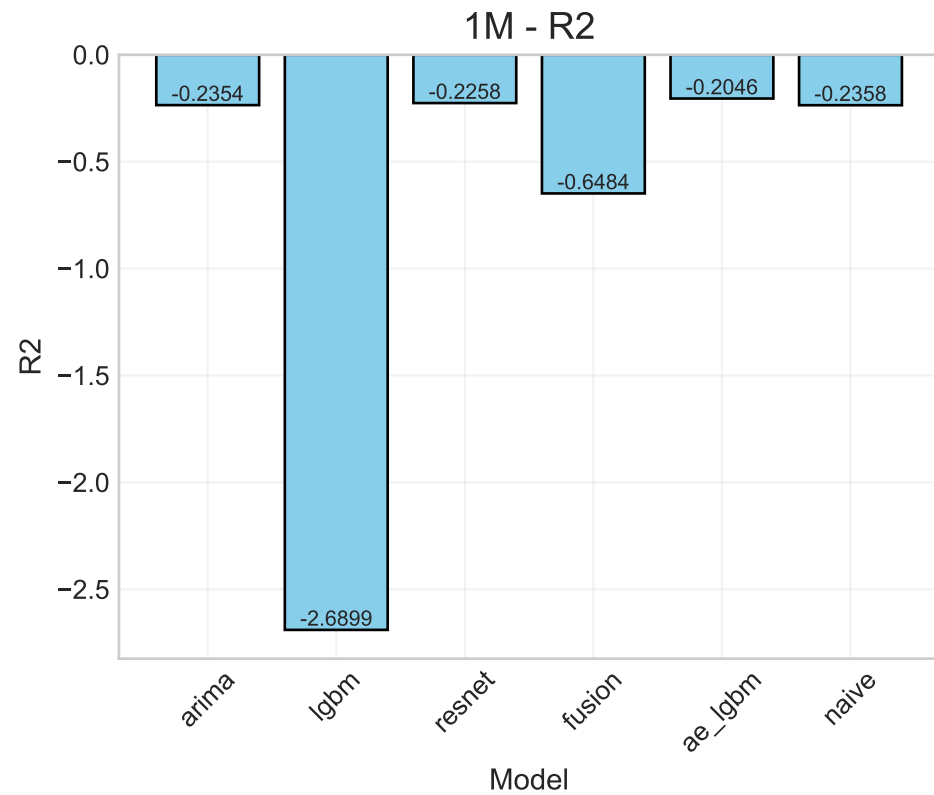
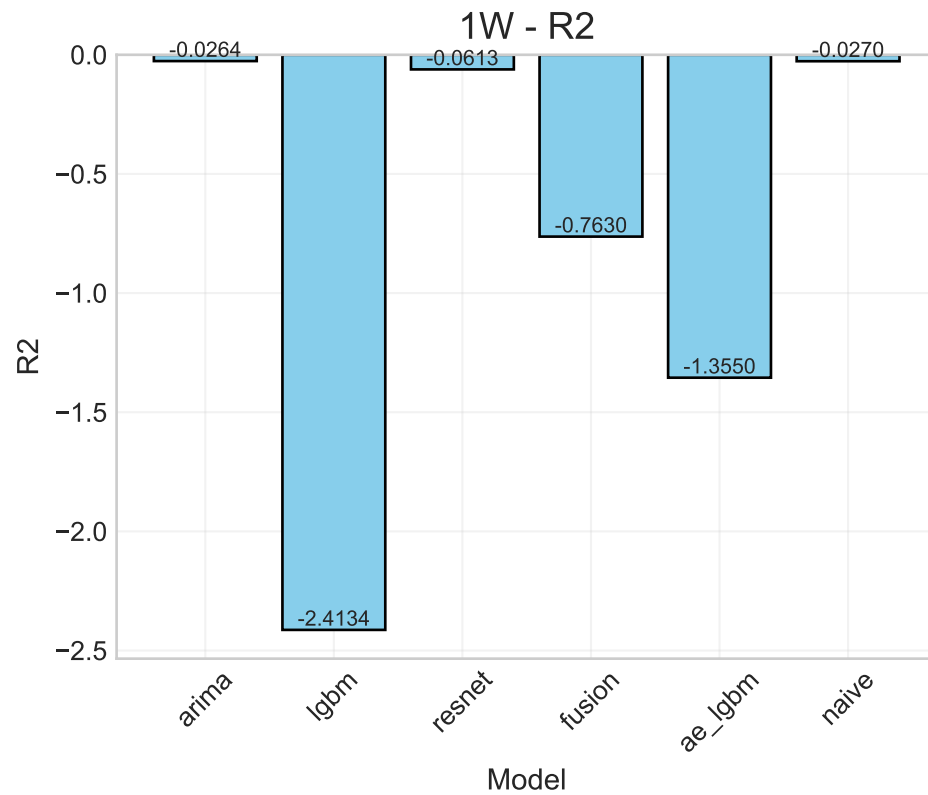
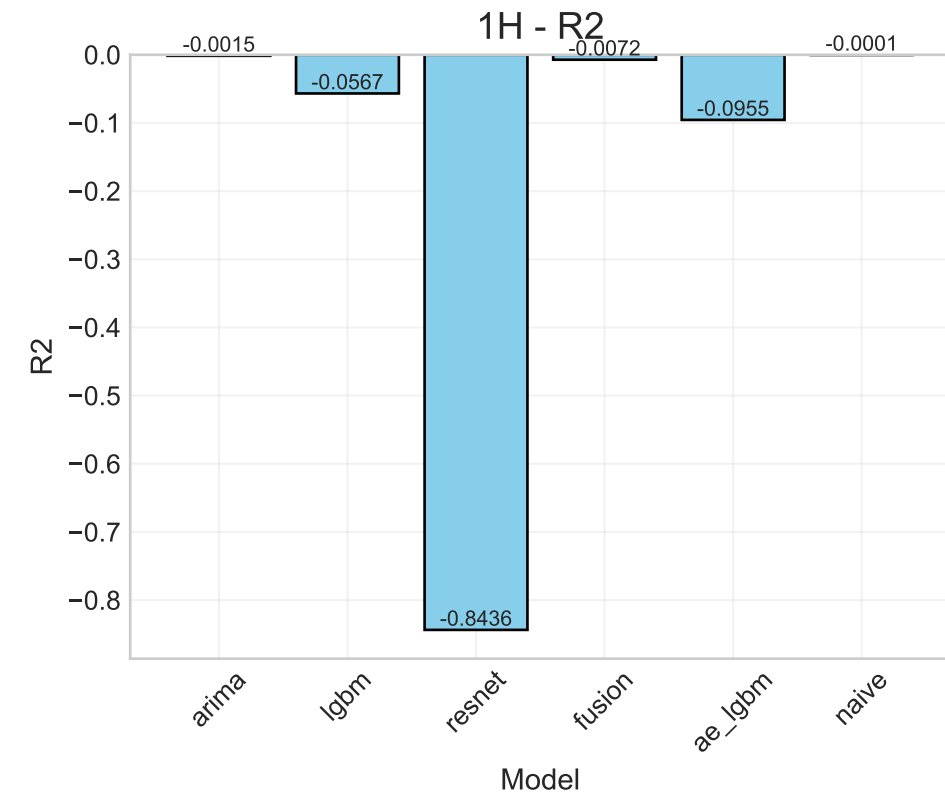
1W - Directional_Accuracy



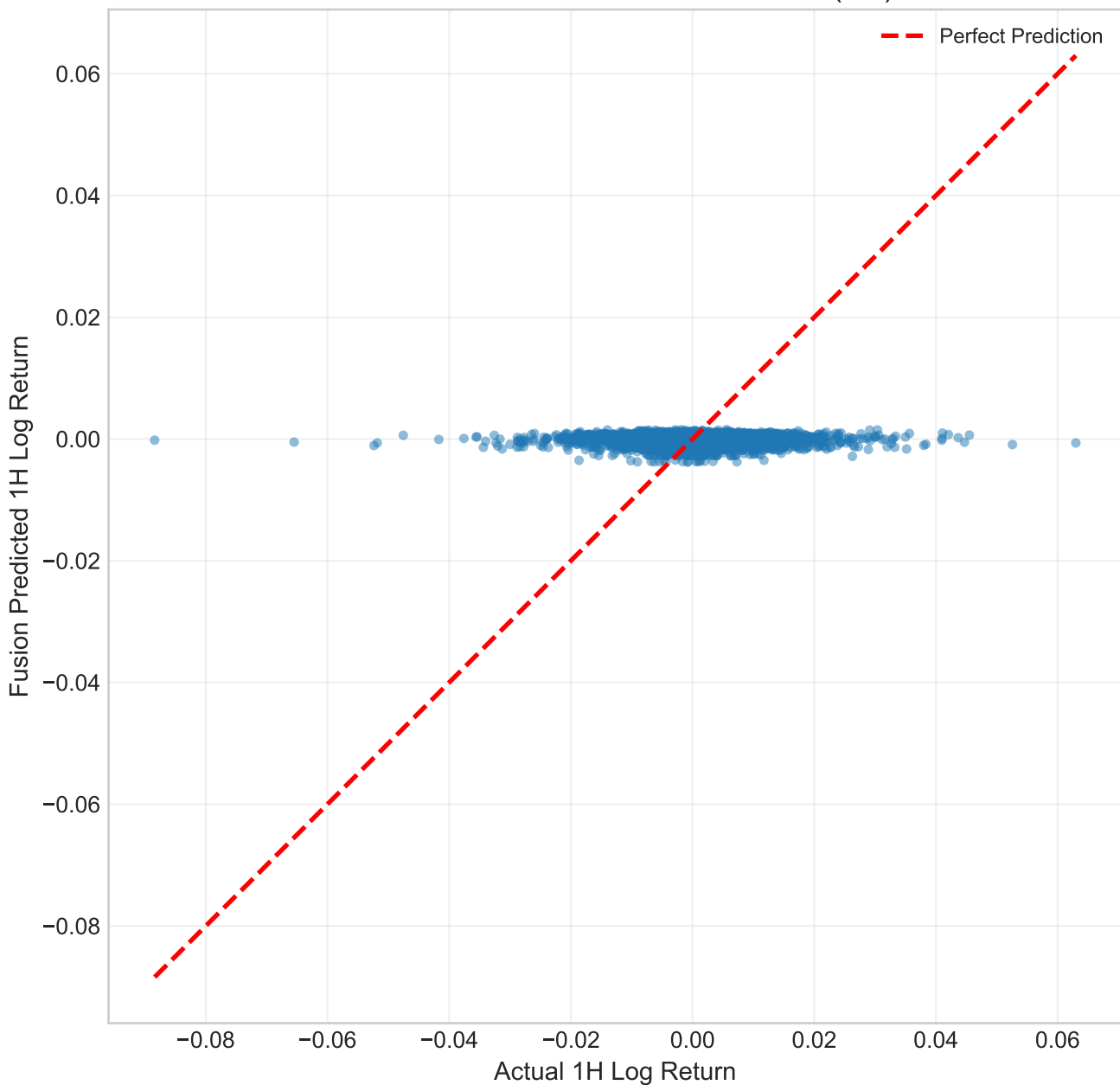
1M - Directional_Accuracy



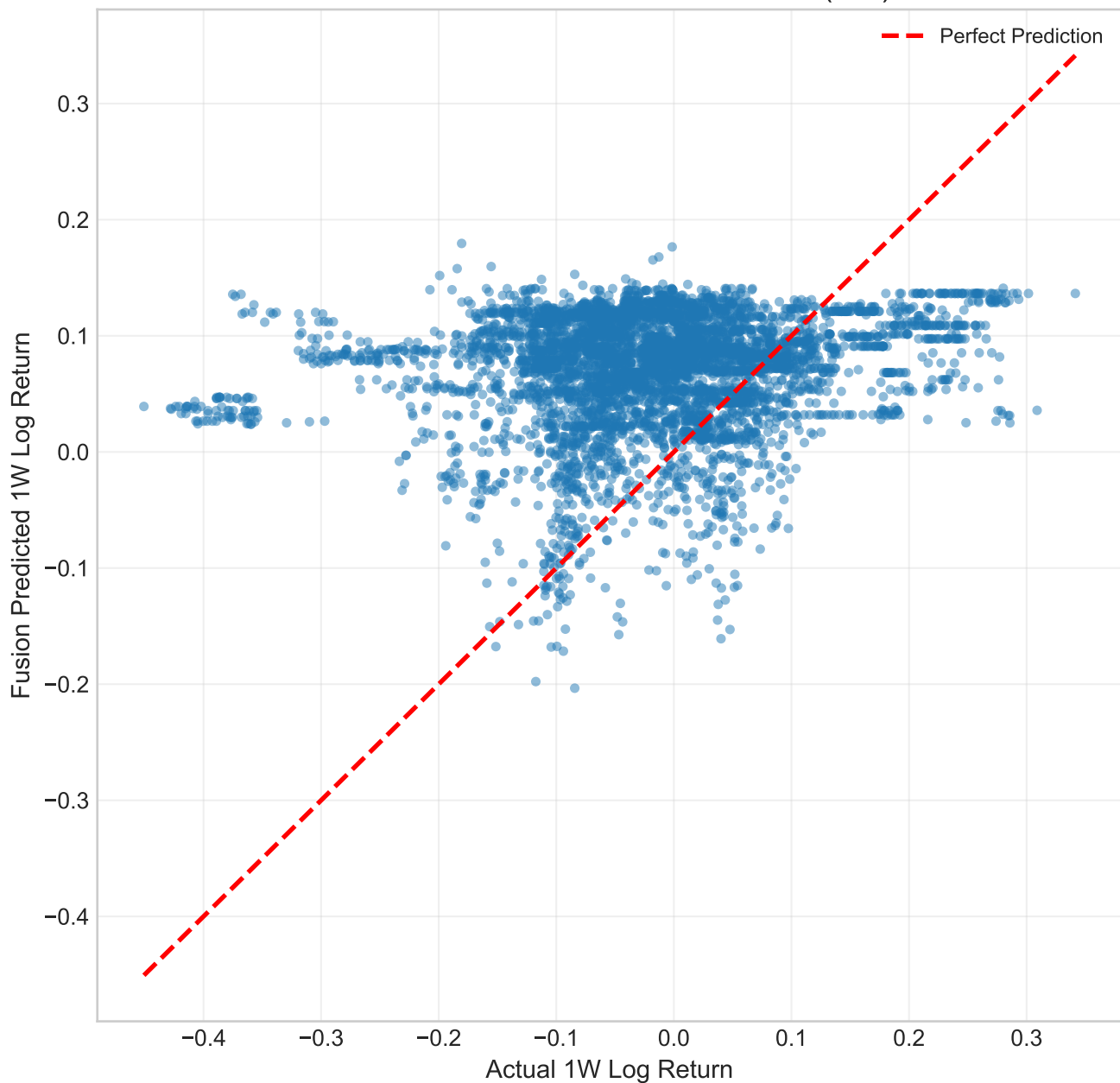
Comparison of R2 Across Models and Horizons



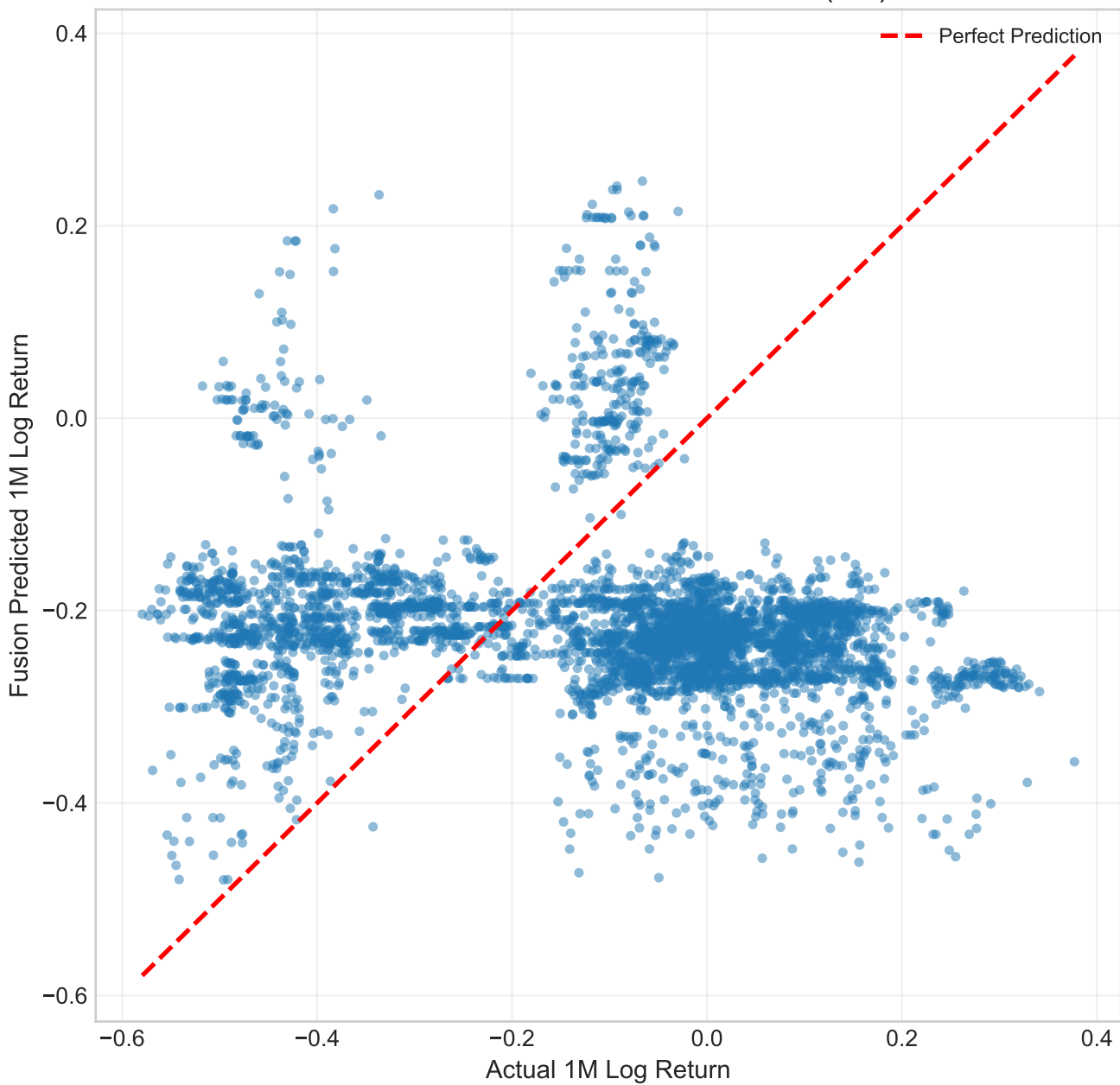
Fusion Model: Predicted vs Actual (1H)



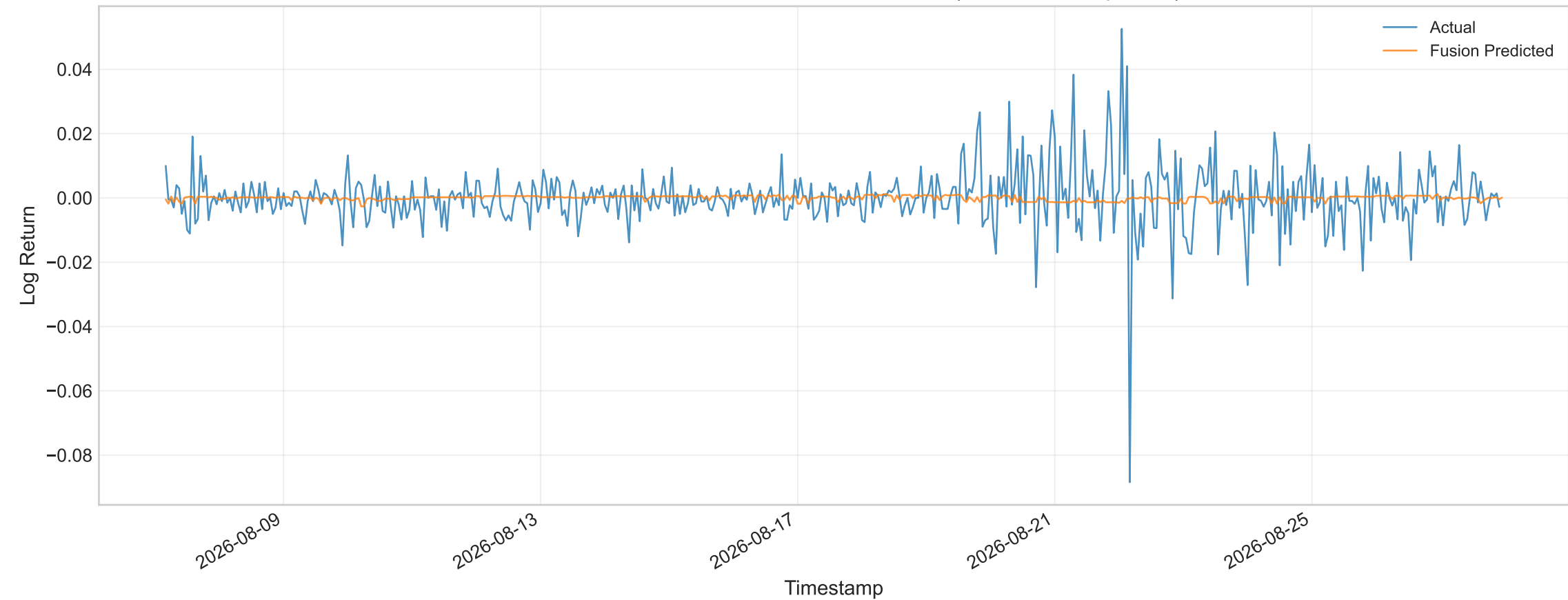
Fusion Model: Predicted vs Actual (1W)



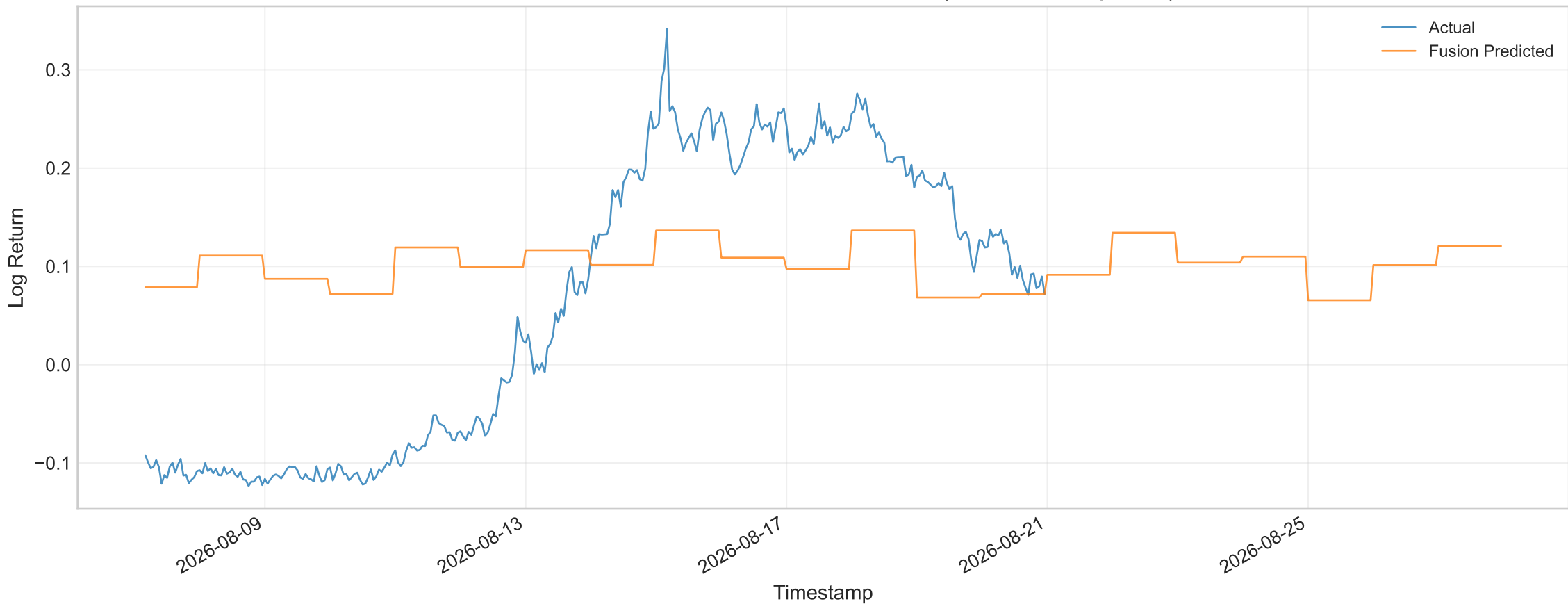
Fusion Model: Predicted vs Actual (1M)



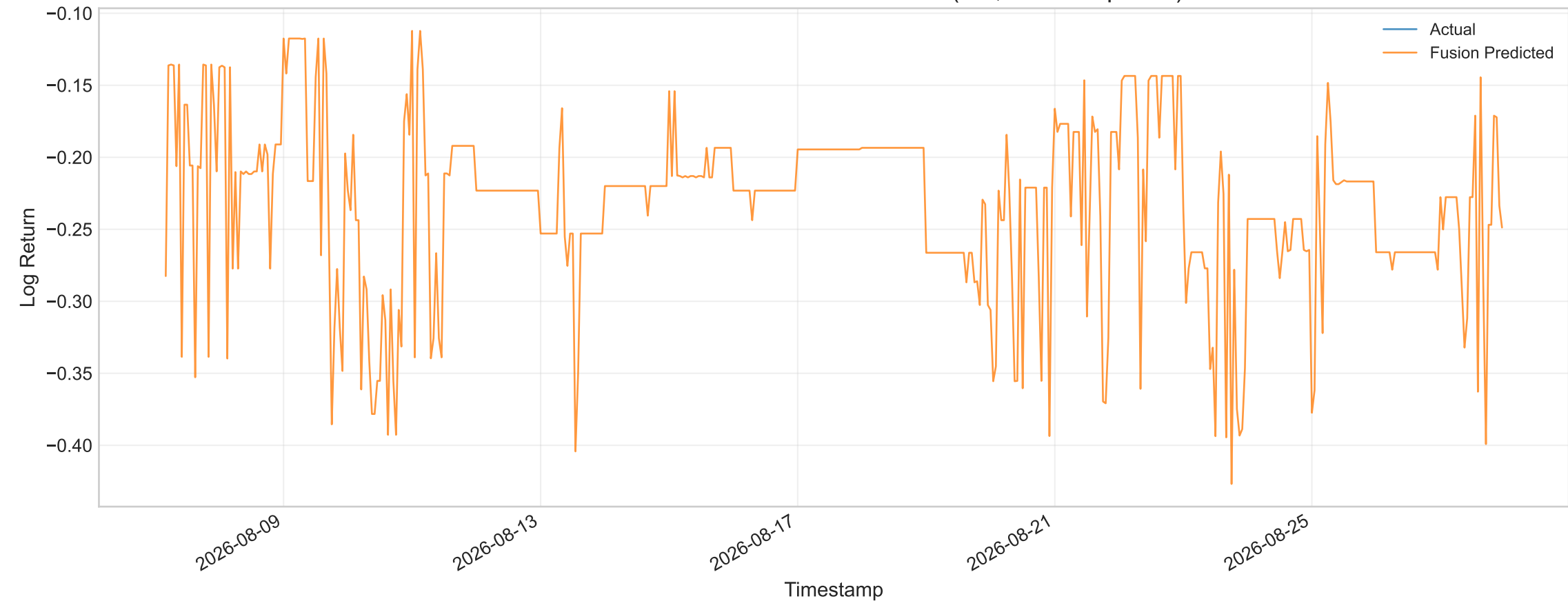
Time Series: Actual vs Fusion Prediction (1H, last 500 points)



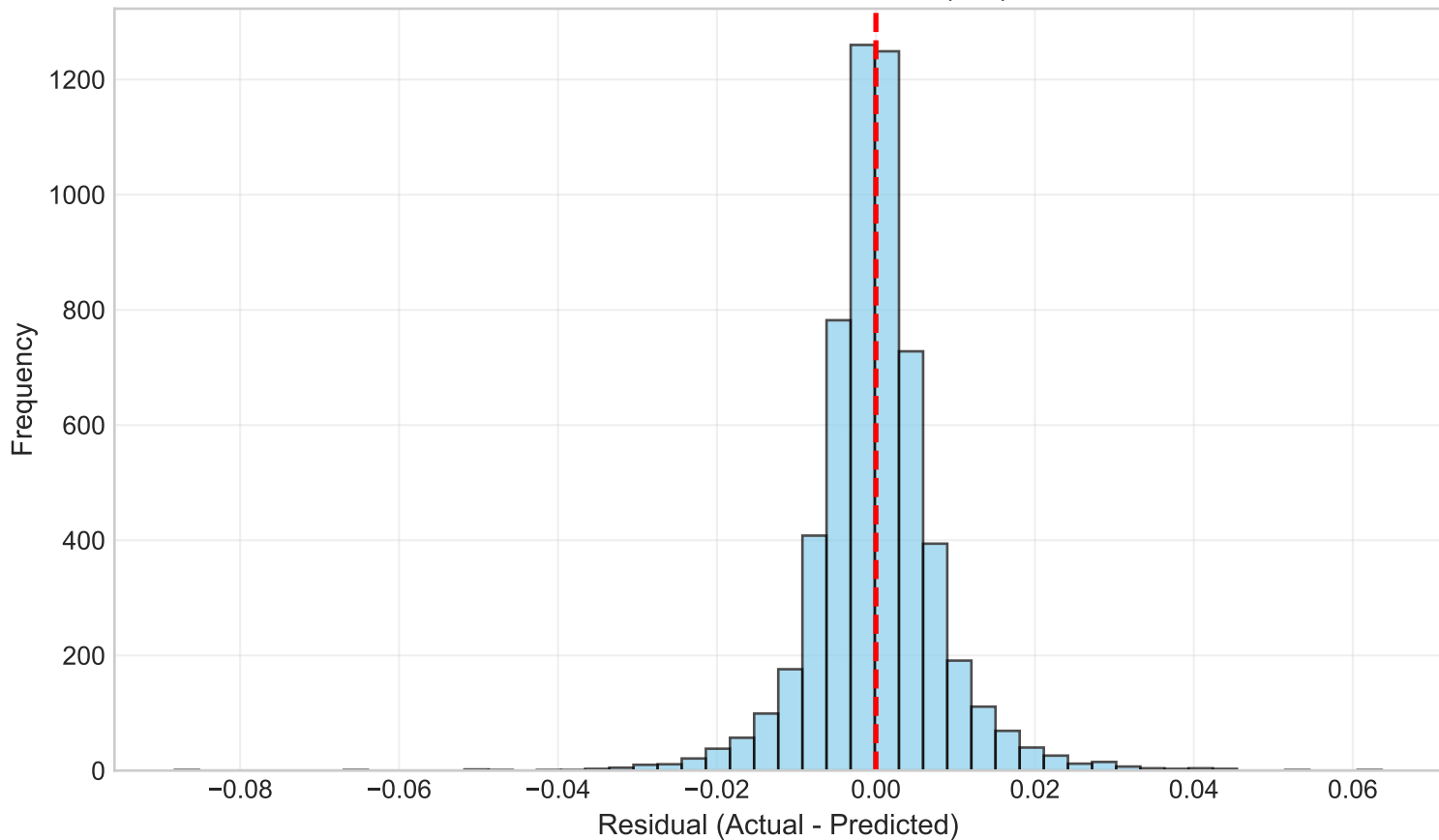
Time Series: Actual vs Fusion Prediction (1W, last 500 points)



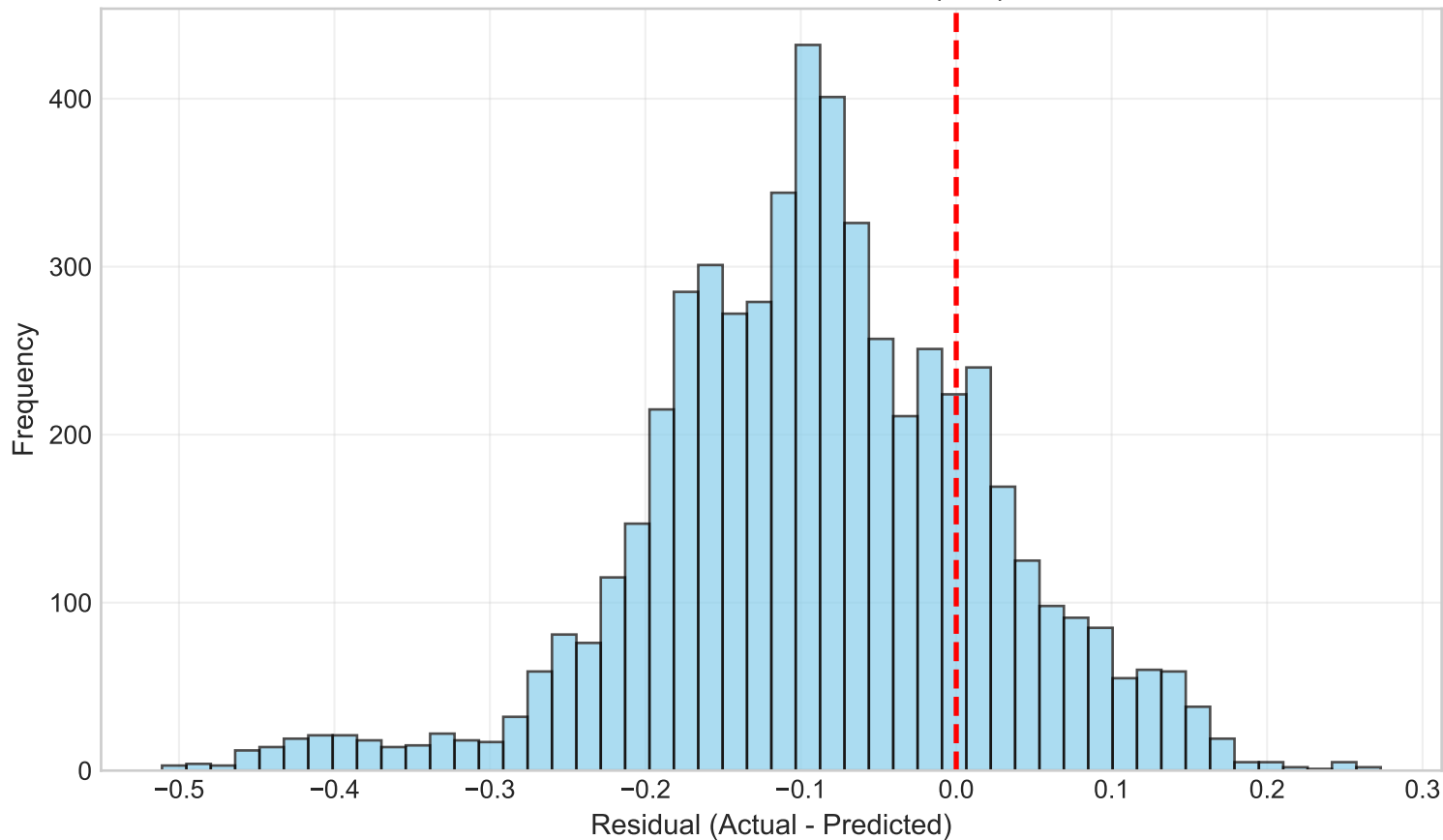
Time Series: Actual vs Fusion Prediction (1M, last 500 points)



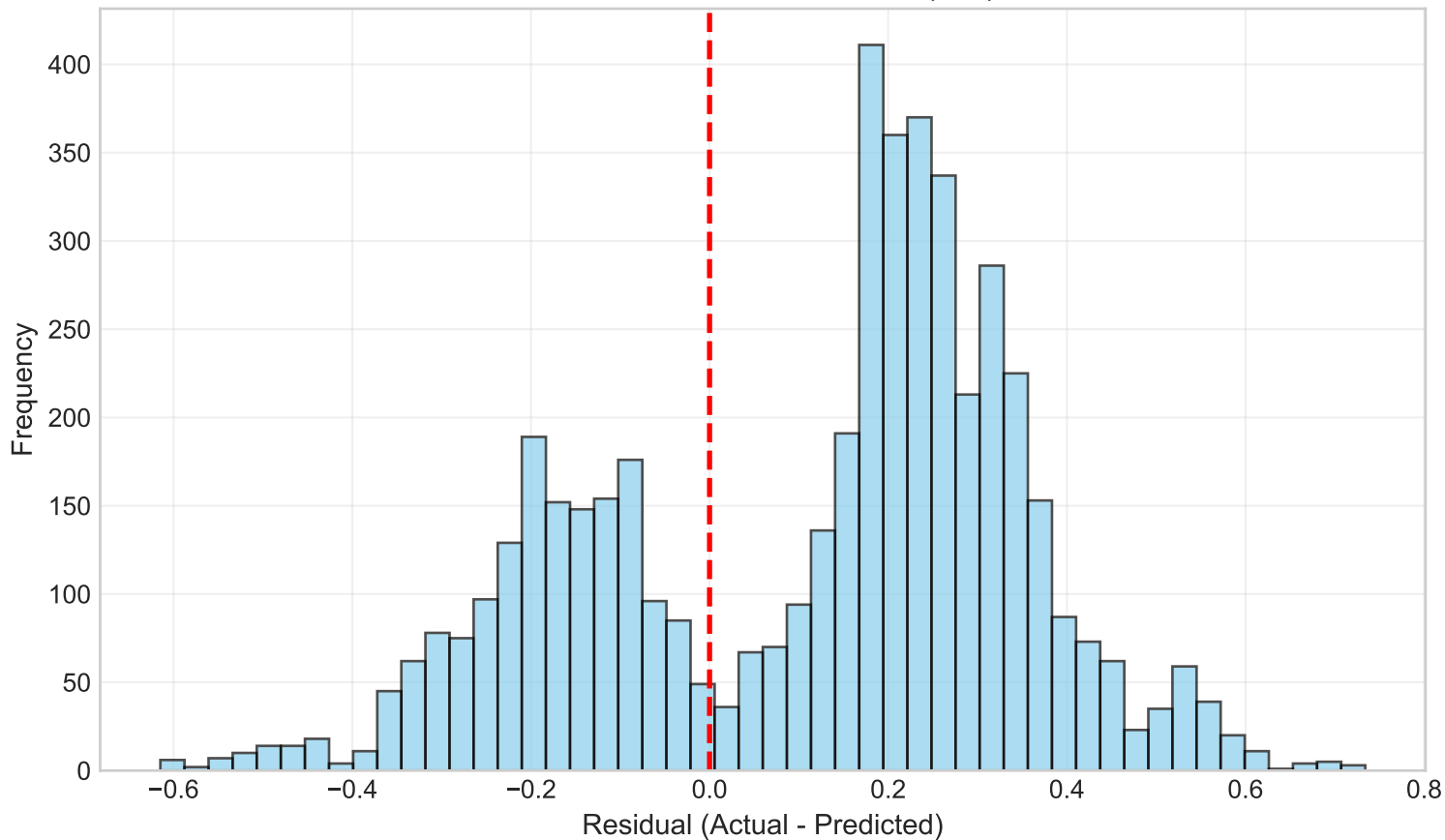
Fusion Model Residuals (1H)



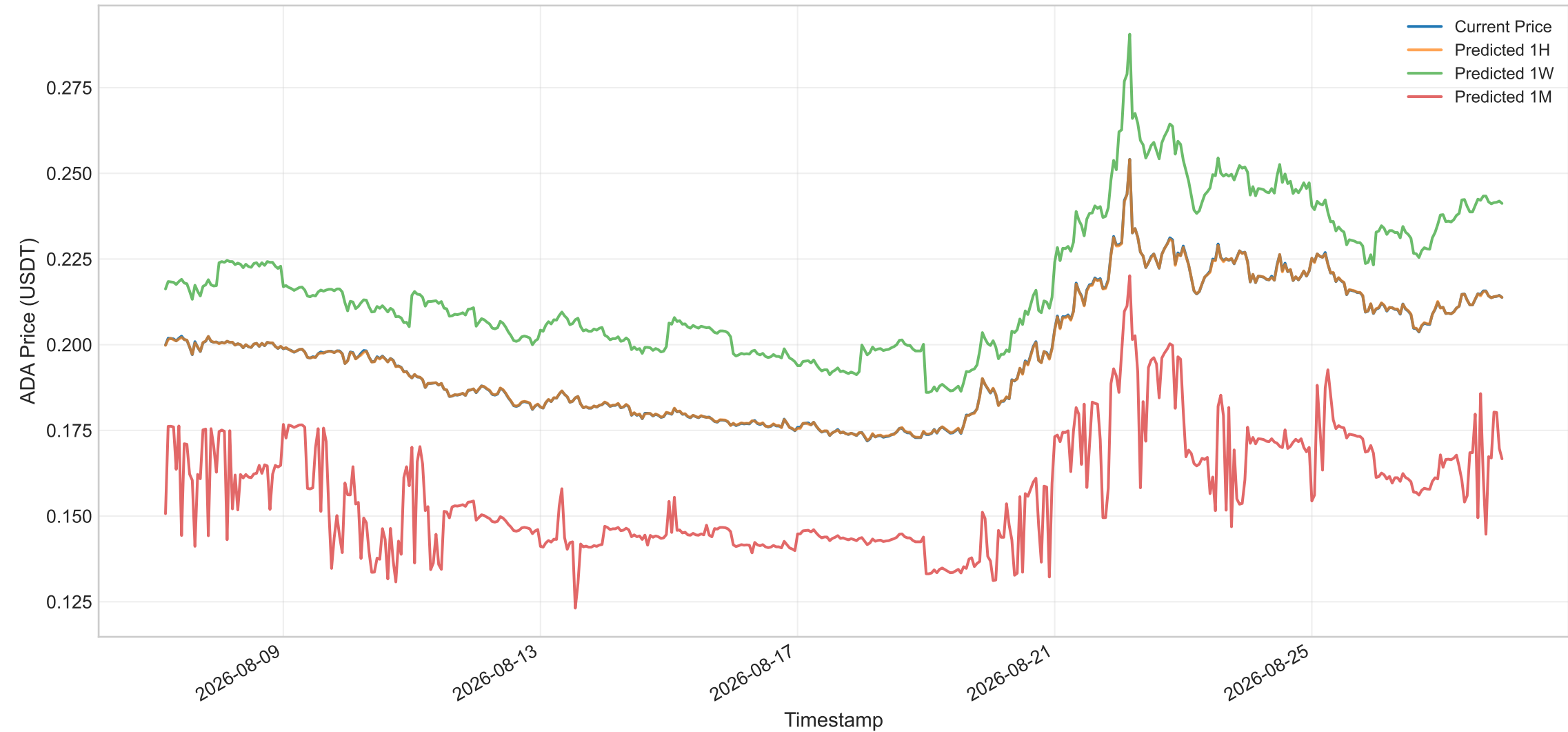
Fusion Model Residuals (1W)



Fusion Model Residuals (1M)



Current and Predicted Prices (last 500 points)



Pipeline Metadata

```
dataset: C:\Users\Ali\Documents\github_alibavarchiee\ADAXXX\ADAUSDt_multiscale_long.csv
asset: ADA
market: ADAUSDt
source: Binance Spot
start: 2022-08-27
end: 2026-08-27
prediction_horizons: ['1H', '1W', '1M']
canonical_frequency: 1H
models: ['ARIMA', 'LightGBM', 'AutoEncoder', 'AE+LightGBM', 'ResNet18', 'Fusion LightGBM']
device: cpu
rows_modeling_dataset: 35087
rows_train: 24935
rows_validation: 4416
rows_test: 5736
```

End of Report